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Submarine canyon sediment transport and accumulation during sea level highstand: Interactive seasonal regimes in the head of Astoria Canyon, WA

E.J. Lahr[a,*], A.S. Ogston[a], J.C. Hill[b], H.E. Glover[c], K.J. Rosenberger[b]

[a] School of Oceanography, University of Washington, Seattle, WA, USA
[b] Pacific Coastal and Marine Science Center, United States Geological Survey, Santa Cruz, CA, USA
[c] College of Engineering, School of Civil & Construction Engineering, Oregon State University, 101 Kearney Hall, Corvallis, OR 97331, USA

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ABSTRACT

The majority of submarine canyons on Earth today do not directly intersect littoral or fluvial sediment sources, yet these systems are rarely studied. The shelf-incised head of Astoria Canyon receives sediment from the nearby Columbia River and is subject to energetic forcing from shelf and slope processes, making it an ideal site to evaluate the modern activity of canyons in high-stand sea level conditions. This study uses in-situ data from Astoria Canyon to identify the active sediment transport processes and patterns of accumulation in temperate canyon systems that are decoupled from their sediment sources during sea level highstand. Hydrodynamic data from a benthic tripod deployment in the head of Astoria Canyon shows that sediment resuspension and transport during summer is driven by internal tides and plume-associated nonlinear internal waves. Observations of shoreward-directed currents and low shear stresses (<0.14 Pa) along with sediment trap data suggest that seasonal loading of the canyon head occurs during summer. Nearby long-term wave data show that winter storm significant wave height often exceeds 10 m, driving shear stress capable of resuspending all grain sizes present within the canyon head. Swell events are generally concurrent with downwelling flows, providing a mechanism for episodic downcanyon sediment flux. Century-scale accumulation rates evaluated from sediment cores show slow accumulation in the upper canyon head, but rates progressively increase with depth to at least 300 m. The depositional environment in Astoria Canyon continues to respond to fluvial and oceanic forcing over an annual cycle. This study indicates that canyon heads can continue to function as sites of sediment winnowing and bottom boundary layer export even with a detached, shelf-depth canyon head.

1. Introduction

Submarine canyons are morphological features present along nearly every continental margin on Earth, and they are sites of significant ocean mixing, sediment transport, and nutrient exchange between coastal and abyssal environments (Carter and Gregg, 2002; Puig et al., 2014; Azpiroz-Zabala et al., 2017; Thomsen et al., 2017; Paull et al., 2018). Many of these submarine canyons were initially incised during the Pleistocene, when terrestrial sediment was supplied directly to the shelf break and frequent sediment gravity flows incised canyon thalwegs (Shanmugam et al., 1985; Piper and Normark, 2009). The frequency of these flows dramatically decreased during the Holocene transgression when the coastal sediment supply was displaced landward of the shelf break, and canyon floors began infilling with hemipelagic sediment (Carlson, 1967; Nelson, 1976). However, episodic density-driven sediment transport processes remain an active process in submarine canyons during the Holocene (Heijnen et al., 2022). Identifying the triggers for both episodic and tidal-to-seasonal transport events is critical for understanding canyon functioning, geochemical fluxes, and continental margin stratigraphy.

Near-bed hydrodynamic observations provide the most detailed evidence for how sediment is transported through the canyon environment, and these types of observations have revealed a wide spectrum of canyon sediment transport processes. Processes that act to incrementally distribute sediment throughout canyons include biogenic pelagic sedimentation, settling from river plumes, internal and surface tides, and subtidal currents (Puig et al., 2014). Punctuated transport events have been initiated by aseismic slope failure (Xu et al., 2013), storm-induced liquefaction (Puig et al., 2004), surface wave shear stresses (Ma et al., 2008; Traykovski et al., 2007), dense shelf-water cascading (Canals

* Corresponding author.
E-mail address: elahr@uw.edu (E.J. Lahr).

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et al., 2006), hyperpycnal river plumes (Warrick et al., 2008; Liu et al., 2012), and hypopycnal river plumes (Hage et al., 2019). The cumulative effect of transport within canyon systems dictates the locations and rates of sediment accumulation, in some cases driving deposits toward unstable morphologies that are prone to mass failure. This preconditioning of canyon sediment can drive sediment gravity flow triggering, and recent studies have found that major canyon-confined turbidity currents do not require major external triggers if there are unstable deposits (Paull et al., 2018; Bailey et al., 2021). Further, the processes responsible for unstable sediment preconditioning are poorly understood, particularly in canyons which are detached from land under high-stand yet actively redistribute sediment.

The regional setting of this study is the northern Cascadia Margin (Fig. 1), where morphology is characterized by a 40 km wide continental shelf, steep continental slope, and abyssal trench which has been infilled by Plio-Pleistocene sedimentation. The Cascadia Margin contains eleven major canyon systems which fit the Pleistocene-Holocene timeframe of incision and infill, evidenced by deep-water fan geochronology (Nelson, 1976; Knudson and Hendy, 2009), seismic profiles, multibeam bathymetry (Hill et al., 2020), and canyon sedimentation (Carlson, 1967, Carson et al., 1986). Turbidites and other sedimentary structures emplaced by sediment transport events punctuate hemipelagic Holocene stratigraphy from submarine canyons and channels throughout the Cascadia Margin (Adams, 1990; Nelson et al., 2000; Paull et al., 2014). Some of these deposits have been attributed to large (7–9 M_w) subduction zone earthquakes, but several coarse and fine layers remain unattributed to triggering events (Carlson, 1967; Goldfinger et al., 2012). Observations of in-situ sediment transport are rarely made within canyons on the Cascadia Margin or elsewhere due to the logistical challenges of data collection. As a result, the balance between event-driven and gradual sedimentation in high-stand canyons remains poorly understood.

The aim of this study is to expand our understanding of resuspension and transport processes in high-stand submarine canyons and identify the processes that deposit and transport canyon-head sediment. We hypothesize that sediment transport patterns during summer load the canyon head with sediment, preconditioning the seabed in the upper channel for episodic remobilization during winter conditions. To address this hypothesis, we combine hydrodynamic time series observations with cores obtained from modern sediment deposits in the head of Astoria Canyon. This combined approach allows for the comparison of canyon hydrodynamic processes and seabed accumulation, connecting modern processes to sediment loading in the canyon head. We evaluate this in situ data in the context of the Cascadia Margin and contrast this high stand canyon with a range of active and inactive systems.

2. Background

2.1. Astoria Canyon

The study site is focused over a 200 km^2 area encompassing the head of Astoria Canyon, one of north Cascadia's major canyons. Astoria

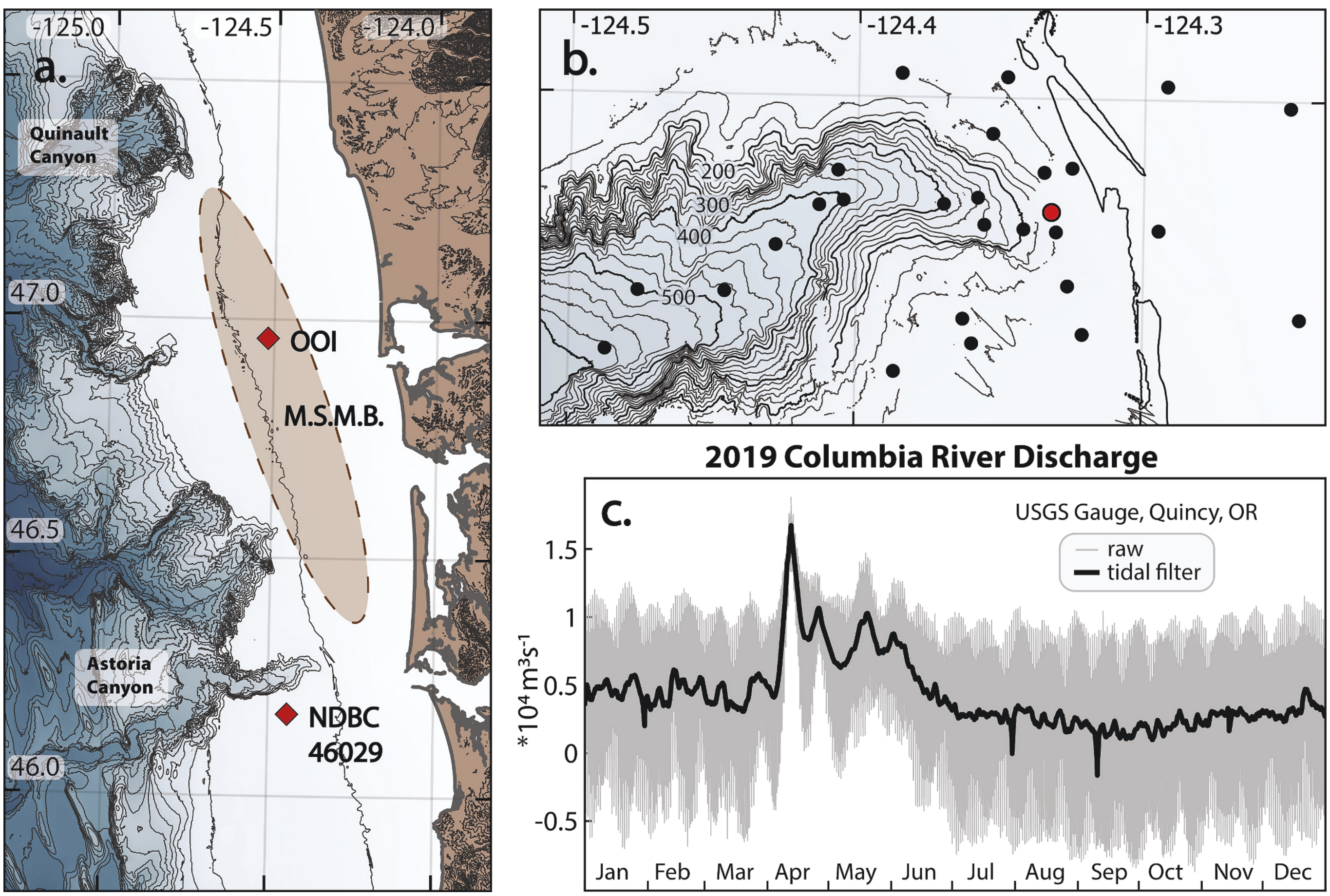

Fig. 1. a. Map of the northern Cascadia Margin, including Astoria Canyon and the mid-shelf mud belt, with bathymetry contours at 100 m intervals, and b. bathymetry at the head of Astoria Canyon. Data sources are noted, including sediment core locations (black circles), the tripod site (red circle), the National Data Buoy Center Station 46,029 and the Ocean Observatories Initiative Washington Shelf Surface Mooring (black diamonds). c. Columbia River hydrograph at Quincy, OR, in 2019 (40-h low-pass filtered data in black, raw data in gray). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Canyon was formed as a conduit for Columbia River sediment during glacial low-stand, and the stratigraphy of Astoria Fan indicates that canyon incision began as early as 2.58 Ma (Bouma and Normark, 1984). Holocene sea-level rise from 20 to 7 ka halted canyon incision (Carlson, 1967), and following the Holocene transgression, fine-grained sediment from the Columbia River aggraded a broad northeast-trending mud deposit on the shelf of up to 14 m thick, referred to as the mid-shelf mud belt (Nittrouer et al., 1979). Over the last century, roughly two thirds of the annual sediment supply from the Columbia River has been retained on the mid-shelf mud belt (Nittrouer et al., 1979; Sternberg, 1986). The southern terminus of the mud belt intersects the head of Astoria Canyon, where canyon bathymetry is believed to capture a significant amount of fine-grained sediment annually, equivalent to ~5 % of the Columbia River sediment load (Sternberg, 1986). On seasonal timescales, fine-grained deposits several centimeters thick have been observed on the shelf during summer, but are absent in winter, and are believed to be eroded by winter storms (Nittrouer and Sternberg, 1981). Beyond the shelf break, the slope and canyon regions accumulate modern sediment more slowly, with a majority of modern sedimentation focused on the upper portion of the slope (Carpenter et al., 1982). At depths greater than 500 m within the canyon axis, coarse sediment layers are commonly found in modern deposits from the canyon thalweg, and intermittent coarse sediment layers persist onto Astoria Fan (Carlson, 1967).

2.2. Sediment supply to the canyon head

The Columbia River is the largest river (670,000 km^2 basin area, 260 $km^3\ y^{-1}$ discharge) and the second largest source of fluvial sediment (8 MT; Naik and Jay, 2011) on the west coast of North America. The hydrograph of the Columbia River peaks in spring and summer in the northern hemisphere (Fig. 1), driven by high-elevation snowmelt, with lesser peaks in winter driven by rainfall. The Columbia River's natural hydrograph has been altered and sediment supply has been greatly reduced by dams and climate change. Nearly all the reduction in sediment supply is in sand-sized particles, while fine-grained sediment supply has remained largely unchanged (Naik and Jay, 2011).

On seasonal timescales, the plume pathway is controlled by wind stresses and along-shelf geostrophic currents (Hickey et al., 1998). Southerly, downwelling-favorable winds during winter force the plume northward along the coast, whereas northerly winds during summer force the plume to travel southwest or bifurcate into north- and southwest-moving branches (Hickey et al., 2005). Plume sediment concentrations are typically less than 30 mg l^{-1} beyond the river mouth (Sherwood and Creager, 1990; Spahn et al., 2009; Nowacki et al., 2012), and the plume front fluctuates tidally.

2.3. Factors impacting sedimentary processes within Astoria Canyon

The surface wave climate in the northeast Pacific Ocean is energetic, with deep-water storm swell annually reaching a significant wave height of 10 m, and interannual maxima reaching 15 m (Allan and Komar, 2002; Allan and Komar, 2006; Ruggiero et al., 2010). These storm events drive strong seabed shear stresses over the continental shelf, capable of resuspending sediment on the outer shelf ~175 h per year (Sternberg, 1986).

The California Current and Undercurrent System flows parallel to regional shelf isobaths over the shelf-incised upper thalweg of Astoria Canyon. Below the canyon rim, steep canyon walls disrupt shore-parallel geostrophic currents, and mean flow is directed approximately east-west along the canyon axis (Hickey, 1997; Bosley et al., 2004). Wind stresses drive upwelling- and downwelling-favorable conditions over the canyon with timescales of 2 to 10 days, and associated currents below the canyon rim have been observed with along-axis velocities of 1 m s^{-1}. Currents as deep as 600 m within the canyon maintain a statistical relationship with wind stresses (Hickey, 1997; Bosley et al., 2004).

Bottom nepheloid layers (50 m thick) and intermediate nepheloid layers (200–300 m depth) are observed in the axis of Astoria Canyon (Hickey, 1994; Bosley et al., 2004). In other canyons globally, the breaking of internal tides along slope and canyon bathymetry is known to elevate near-bed shear stress, resuspend seafloor material, and mix sediments high into the water column (Gardner, 1989; Carter and Gregg, 2002; Kunze et al., 2002; Boegman and Stastna, 2019; Maier et al., 2019), contributing to nepheloid layer formation. Tidal currents and internal tides (ITs) are present within Astoria Canyon, amplified by the bathymetry, and nonlinear internal waves (NLIWs) have been observed along the Washington and Oregon shelf (Nash and Moum, 2005; Moum et al., 2007; Alford et al., 2012; Zhang et al., 2015).

3. Methods

3.1. Hydrodynamic data collection and analysis

A bottom boundary layer tripod was deployed into the head of Astoria Canyon at 46.27 N 124.33 W in a nominal water depth of 160 m during the summer of 2019 to observe canyon processes near the annual peak in discharge from the Columbia River. The tripod was deployed on 5 May 2019 and recovered on 12 August 2019. Tripod instrumentation quantified the wave, current, and turbidity parameters relevant to bottom boundary layer sediment transport, as well as temperature and salinity (see Table 1 for details). To understand these values in an annual context, tripod data was supplemented with a year-long record of wave heights and near-bed currents. Supplementary data were obtained from a National Oceanic and Atmospheric Administration buoy (NOAA National Data Buoy Center 46029, 2019), and from the Ocean Observatories Initiative (OOI, Ocean Observatories Initiative, 2019) Washington Shelf Surface Mooring. Both instruments supply concurrent spectral surface wave and meteorological data, and the OOI mooring also recorded near bed current velocity for a complete year (the year 2016 was chosen for its complete records; details in Table 2). The NDBC buoy was located 10 km south of the tripod site, and the OOI mooring was located 81 km north of the tripod site (Fig. 1).

For the summer, seafloor shear stresses were calculated using hourly bursts from the 5 MHz Acoustic Doppler Velocimeter (ADV) mounted 90 cm above bed. First, average current speed was calculated from each hourly current data ensemble, and wave statistics were calculated from each ensemble using PUV wave analysis (more details in Table 2). For the winter, wave statistics and near-bed average velocity were calculated from the OOI mooring. Combined wave-current shear stresses were then calculated for all time periods using the Grant-Madsen wave-current interaction model (Madsen, 1995; Wiberg and Sherwood, 2008).

Optical backscatter sensor data were calibrated from nephelometric turbidity units to units of suspended sediment concentration (mg l^{-1})

Table 1
Instrument deployment details for the canyon head tripod, including sampling frequency and the height above bed of the sampling volume.

Instrument	Sampling Strategy	mab
RDI 300 kHz ADCP, upward-facing	1 burst / min (10 profiles, 0.5 Hz)	2
Sentinel V 1 MHz ADCP, downward-facing	1 burst / 5 min (90 profiles, 0.5 Hz)	1.3
Sontek 5 MHz ADV	1 burst / 60 min (4800 samples, 16 Hz)	0.9
Seabird SBE-37 Microcat T-S Sensor	1 burst / 60 min (4800 samples, 16 Hz)	1.05
RBR Concerto CTD	1 sample / 5 min	1.35
Campbell Scientific OBS 3+	1 sample / 5 min	0.25, 1.9
Sea-Tech Transmissometer	1 sample / 5 min	1.35
RBR Virtuoso (Turbidity)	1 sample / 30 min	1.55
Tube Sediment Trap		0.35, 1.9

Table 2
Input data and variables for the Grant-Madsen wave-current interaction model (Grant, 1979) used to predict bed shear stress within Astoria Canyon head.

Symbol	Name	Summer	Winter
U_{br}	Representative wave-orbital velocity	5 MHz ADV velocity ensemble (PUV method)	OOI Spectral Wave Data (ubspecdat method; Wiberg and Sherwood, 2008)
Ω_r	Representative angular wave freq.	5 MHz ADV velocity ensemble (PUV method)	OOI Spectral Wave Data (ubspecdat method; Wiberg and Sherwood, 2008)
U_c	Current velocity	5 MHz ADV velocity ensemble, hourly average	OOI ADCP near-bed velocity (lowest cell in velocity profile)
Z_r	Reference height for current velocity	0.9 m	4 m
K_n	Bottom roughness height	3 mm	3 mm

using the methods of Ogston et al. (2000). The sensors were placed in a tank with a magnetic stir plate, and subjected to incrementally increasing suspended sediment concentrations (SSC) while sensors recorded data. Each time sediment concentrations were increased, a water sample was collected, and suspended sediments were filtered, dried, and weighed. Sensor output was calibrated to the mass concentration of suspended sediments using a simple linear fit through the origin and reference values. The sediment used for calibration was collected from the seafloor at the tripod site on the date of recovery. The RBR Virtuoso turbidity sensor provided the best-fit calibration to Astoria Canyon sediments, with an R^2 fit of 0.99.

3.2. Sediment core collection and analysis

In August 2019, 24 60-cm box cores were collected from the canyon head thalweg, walls, and surrounding continental shelf, in water depths ranging from 60 to 400 m. Cores were sub-sampled in the field at 2 cm resolution. In the lab, slabs were x-radiographed and further sub-sampled at depth increments of 1 cm. Disaggregated grain size was measured for all samples. Prior to grain size analysis, samples were dispersed using a sodium metaphosphate solution and a sonication bath. Grain size distribution was measured using a Beckman-Coulter LS13–320 Particle Size Analyzer. The sampler completed three measurement periods which were averaged for the final reported distribution. Each sample's individual volume % measurement varied by <10 %, consistent with calibration distributions demonstrating size distribution recovery to within 2 % of each size class.

Sediment accumulation rates were obtained for each core via ^{210}Pb geochronology using the methods of Nittrouer et al. (1979). Dry sediment samples were spiked with 1 ml of ^{209}Po (10 dpm/ml), and radioisotopes were leached from each sediment sample using 15.8 N HNO_3 and 6 N HCl. ^{209}Po and ^{210}Po were then plated onto a silver planchet, and the decay of polonium isotopes was measured using an Ortek Alpha Ensemble alpha spectrometer.

Supported levels of radioactivity were assumed to be equal to the radioactivity found at the base of cores beyond the layer of logarithmically decreasing radioactivity, and this value was subtracted from each sample's total activity. Where core penetration exceeded the layer of logarithmic decrease, samples had naturally supported activities of ~0.5 dpm g^{-1} on the shelf and ~ 3 dpm g^{-1} in the canyon, and these values are consistent with previous studies (Nittrouer et al., 1979; Carpenter et al., 1982). These values were used for cores that did not penetrate beyond the layer of logarithmic decrease, as slight non-constancy in levels of supported radioactivity have minimal effect on the calculation of accumulation rate. The decrease of unsupported, or excess ^{210}Pb activity with depth in each core was fit with an exponential regression and used to calculate a vertical sediment accumulation rate in cm y^{-1}. The conversion to mass accumulation in mg cm^{-2} y^{-1} was calculated by multiplying the vertical sedimentation rate by the dry-bulk density of each sample.

4. Results

4.1. Hydrodynamics

The canyon-axis currents measured during summer 2019 covaried primarily with nearby wind stresses at NDBC 46029 and tides (Fig. 2, Supplementary Fig. 1.). Velocities ranged from −70 to 70 cm s^{-1}, where negative velocities denote currents flowing to the west (downslope) along the canyon axis. Peak near-bed shear stresses were driven by downwelling-associated downslope flow (e.g., Fig. 2a, year days 130–140, 163–166, 177–182), with near-bed current velocity up to 70 cm s^{-1} and shear stresses as large as 0.14 N m^{-2}. Down-canyon flow events were also accompanied by a decrease in near-bed salinity at the canyon head, and an increase in temperature as large as +0.8 °C. Throughout the time series, SSC covaried weakly with shear stress and was elevated during periods of concurrent upwelling and internal wave activity (e.g., year days 143–148, 150–155, 160–163, Fig. 2b). SSC was not significantly elevated during periods of strong down-canyon flow, e.g., Fig. 2 events at year days 135, 165, 180. Up-canyon flow during internal wave activity was associated with increased salinity and decreased temperature.

During summer, significant wave heights were typically 2–3 m with a dominant period averaging 8–10 s. Moderate storm events in spring increased dominant swell height to 5 m (Fig. 2) and period to 16 s. The canyon-head shear stresses generated by waves of this size were < 0.05 Pa. The nearby OOI datasets provided a means to calculate bed stress throughout the year, and shear stress magnitudes for the summer season are in agreement, with weak wave shear stresses at 160 m depth (Fig. 3a), and current shear stresses consistent in magnitude to those observed in the canyon head (Fig. 3b). Stresses of this magnitude are not capable of resuspending most of the grain size distribution at the tripod site (Fig. 3d). However, shear stresses at the 160 m depth canyon head site predicted from OOI wave and current data during winter increased up to 1.18 Pa, which could resuspend the complete grain size distribution present at the tripod site (Fig. 3c, d). Stresses capable of resuspending all grain sizes at the tripod site were estimated to occur roughly 48 h per year (Fig. 3d).

In addition to surface gravity waves, internal tides and nonlinear internal waves (NLIW) were observed in the ADCP time series data (Fig. 4). These events were identified by strong velocity signatures over short (< 30 min) timescales. When peak tidal currents coincided with upwelling favorable conditions there was also evidence of sediment resuspension near the tripod site (Fig. 5). The onset of elevated SSC was concurrent with a sharp transition from down-canyon to up-canyon flow, and SSC remained elevated (up to 150 mg l^{-1}) for several hours, including during the downslope phase of the internal tide. A rapid drop in temperature (−0.7 °C in 3 h) and increase in salinity was also associated with the upslope phase of the internal tide, and both parameters slowly returned to ambient levels during downslope flow over 12 h. The local bed shear stress and current speed remained low (<0.05 Pa and 30 cm s^{-1}) over the entire tidal cycle.

There were 52 solitary waves and a similar number of NLIW packets identified in the three month tripod deployment. All the waves were low-mode waves of depression which affected the water column over a period of approximately 10 min. An average amplitude of NLIWs of ~30 m was estimated using ADCP backscatter. These waves most frequently occurred at or slightly after low tide and were associated with elevated SSC (10–15 mg l^{-1}). The largest SSC associated with NLIWs was 130 mg l^{-1} during the summer observation period. The direction of NLIW propagation can be identified using current orientations during wave passage: mode-1 depression waves drive opposing deep and shallow currents, and wave propagation is in the direction of the shallow

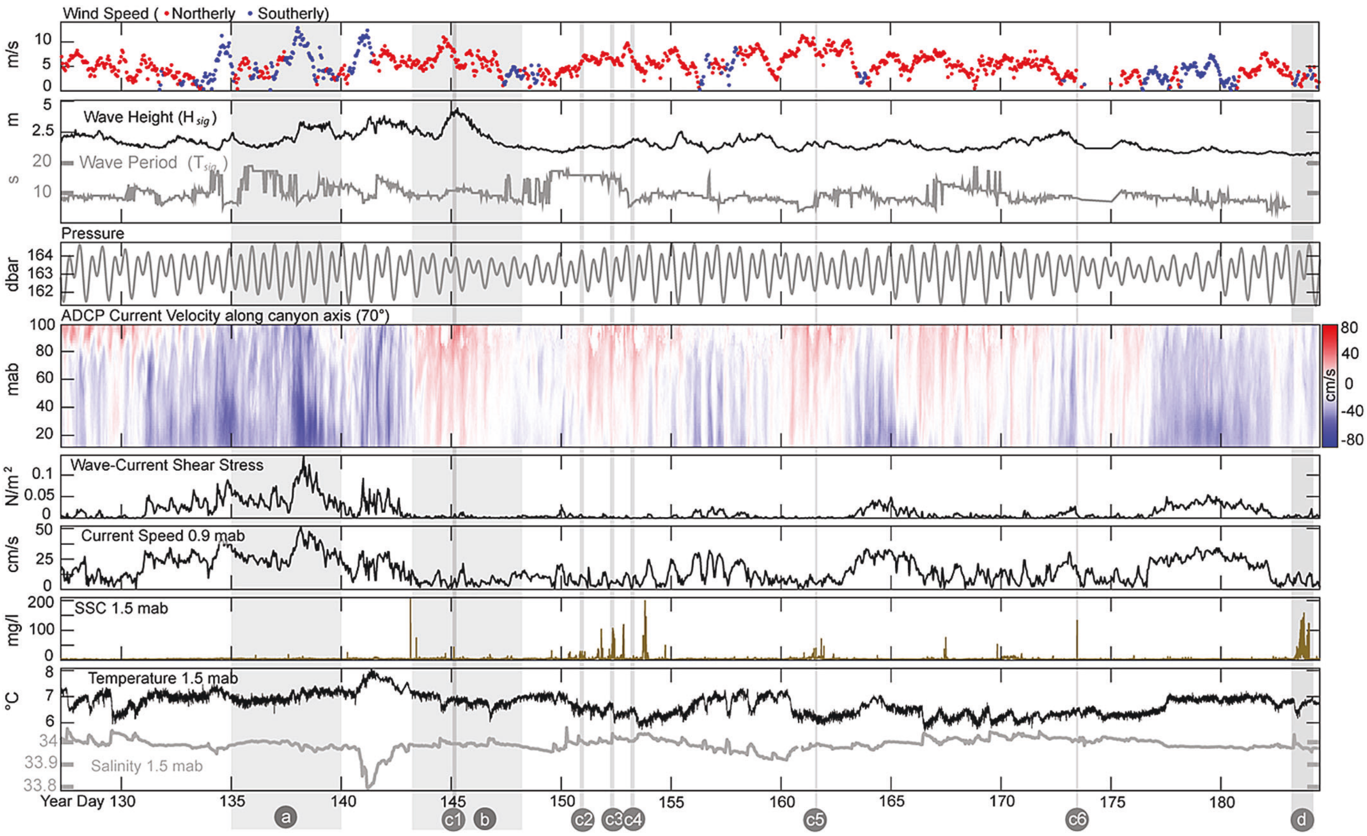

Fig. 2. Time series from the head of Astoria Canyon between 5 May 2019 and 3 July 2019. Wind and wave statistics come from NDBC 46029 and are plotted in red if coming from the north (upwelling-favorable), and blue from the south (downwelling-favorable). ADCP velocity data is plotted in its principal velocity component (which corresponds to the canyon axis), positive values correspond to up-canyon currents. The time periods shaded in gray are expanded in Figs. 4 and 5. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

current. The NLIWs observed within the canyon head drove near-bed currents to the east, indicating wave propagation west from the shore toward the canyon. In a concurrent synthetic aperture radar image, we document a radial, mode-1, Columbia River plume-associated NLIW near the tripod site (Fig. 6).

4.2. Sedimentology

Average seafloor grain size in the canyon thalweg was medium silt, with 73 % silt, 16 % clay, and 11 % sand by volume (Fig. 7). Thalweg sand fraction steadily decreased from 36 % to 6 % downcanyon. There was minimal grain-size variability down-core at the thalweg sites. Sand-fraction was dominant along the shelf surrounding the canyon head (70 %, Fig. 7). Outcrops of consolidated gray clay were cored along the southern rim. Seafloor grain size was similar along the northern shelf rim, with silt and clay increasing with proximity to the canyon walls. Shelf grain size usually decreased slightly down-core. The two sediment traps placed at 0.35 and 2 m above bed collected sediments of identical grain size distribution. The grain size distribution of trap sediment was similar to seafloor samples from the canyon thalweg in deeper water (Fig. 7).

X-radiography of sedimentary structures (Fig. 8) showed extensive bioturbation. Physical structures were overwritten in half of the cores. Unaltered physical layering was preserved in core 22 located north of the canyon head. The physical layers which were preserved consist of coarse silt and fine sand. Shell hash and wood were present in cores 17, 18, 19, 21. The surface mixed layer in the study area varied from 1 to 5 cm below seafloor.

^{210}Pb accumulation rates from the upper canyon thalweg had an average value of 0.67 g cm^{-2} yr^{-1}. Within the canyon head, core accumulation rates increase with water depth to 550 m depth (Fig. 9). Accumulation rates from cores along the north shelf were spatially consistent and averaged 0.24 g cm^{-2} yr^{-1}. Accumulation rates along the south shelf were more spatially variable. The surface mixed layer ranged from 1 to 6 cm below seafloor. A majority of R^2 values for log-linear trendlines were > 0.8 indicating interannually consistent rates of accumulation.

5. Discussion

Observations of sediment transport, grain size and accumulation rates in the head of Astoria Canyon reveal the importance of interacting seasonal sediment supply and ocean energetics in high stand canyon heads. In particular, both winter and summer hydrodynamic processes control sediment transport and accumulation through Cascadia Margin canyons. Previous studies of similar canyons have emphasized the role of winter storms in driving down-canyon transport (Puig et al., 2004; Thomsen et al., 2017), and processes within Astoria Canyon are consistent with these studies. Storm events produce bed shear stresses that can remobilize the full distribution of seafloor grain sizes, not only on the continental shelf, but also within the canyon head over multiple days each year (Fig. 3), and the strong nearbed down-canyon flows associated with southerly winds typical of winter are among the strongest currents in the canyon (Fig. 2).

While the importance of winter storms for canyon sediment transport has been emphasized in many studies (e.g., Meiburg and Kneller, 2010; Puig et al., 2013; Normandeau et al., 2020), the summer tripod data provide new insight into the importance of persistent and less-energetic summer events on loading and retaining sediment in the canyon head. During the upwelling favorable conditions characteristic of summer,

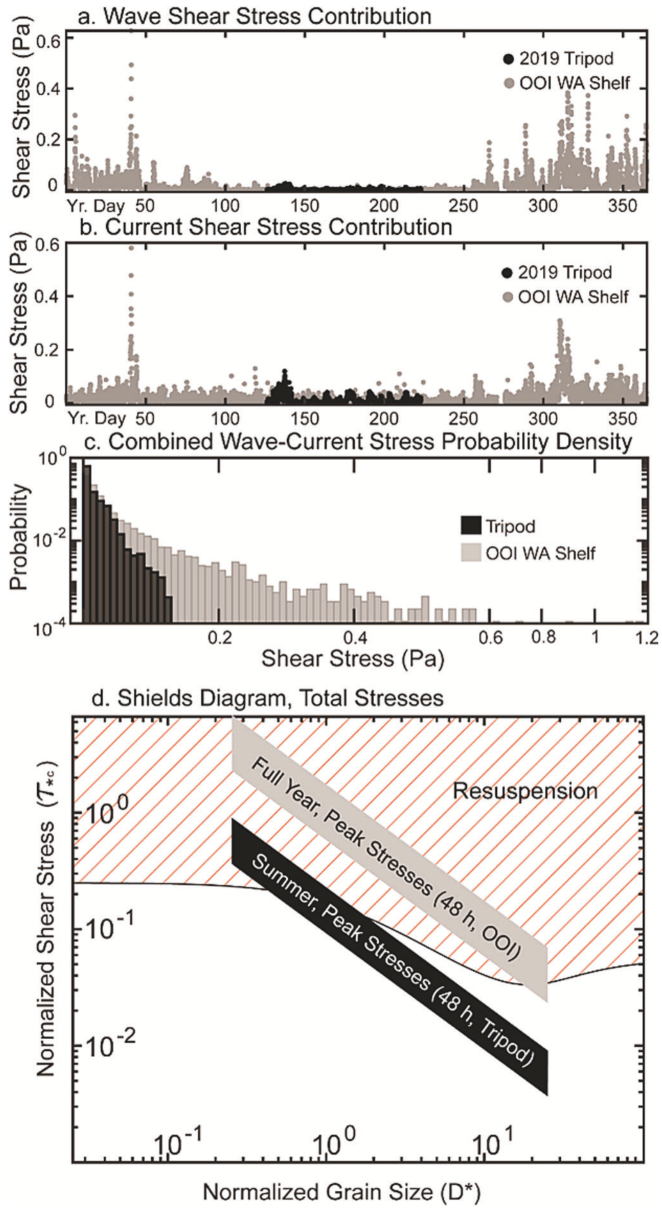

Fig. 3. Shear stress computed from the Astoria Canyon tripod deployment (black) and the OOI Shelf mooring (gray). Panels a) and b) depict relative stress contributions from waves and currents respectively, c) the distribution of total stresses, and d) maximum shear stresses from summer and winter on a Shields diagram (Guo, 2020).

internal waves are prevalent, and suspended sediment is periodically transported landward from mid-canyon depths (Fig. 2). This upwelling favorable state provides an important setting for seasonal plume delivery and retention within the canyon head. Although strong internal-wave currents are short-lived, these resuspension events can slow the consolidation process of the recently delivered plume sediment resulting in high porosity, unconsolidated deposits on the shelf (Nittrouer and Sternberg, 1981) and in the canyon head (Fig. 5). This preconditioning results in a system with a readily available seabed for subsequent winter storm events to flush sediment down the canyon. Understanding and predicting the magnitude of these loading processes is critical to predicting total down-canyon transport, and studies focused on winter extremes would not capture this fundamental link in the sedimentological cycle.

5.1. Summer sediment transport via internal tides and river plume NLIWs

Internal waves play an important role in shaping the morphology of shelf-edge mud deposits (Boegman and Stastna, 2019). The Cascadia Margin exhibits both an active internal wave climate and a large sediment supply, making it a rich site for studies of sediment resuspension and transport by internal waves. The onshore, upslope phase of the internal tide has been identified as a mechanism of sediment resuspension and transport in many canyon and shelf environment globally, and the data from Astoria Canyon support this model and emphasize the seasonal nature of transport processes influencing longer term accumulation. Studies from canyons including Baltimore Canyon (Gardner, 1989), and Halibut Canyon (Puig et al., 2013) document elevated SSC associated with the onset of a cold-water bore moving up-canyon, and persistently elevated SSC on the bore's gradual, downslope phase. This pattern initially transports resuspended sediment upslope, contrary to the typical model of exclusively down-canyon sediment transport. Sediment-trap grain size distributions collected at the tripod site resemble those of the thalweg down-canyon (Fig. 7) indicating that internal tides initially transport sediment from deposits deeper within Astoria Canyon. However, prior studies suggest that sediment returns downcanyon along isopycnal surfaces, generating intermediate nepheloid layers (Gardner, 1989), as have been observed in water-column studies in Astoria Canyon (Hickey, 1997). Internal tides thus play an apparently large role in summer sediment transport within Astoria Canyon and deliver sediment to the canyon head from deeper regions of the canyon over short timescales.

In addition to internal tides, summer sediment transport in Astoria Canyon is directly linked to the Columbia River plume via NLIWs. The canyon head setting near the outer shelf is similar to that of other studies which document resuspension by NLIWs of depression (Quaresma et al., 2007; Zulberti et al., 2020), however the NLIWs observed during summer in Astoria Canyon are generated by the Columbia River, not by remote sources on the continental slope, as confirmed by SAR imagery (Fig. 6; Nash and Moum, 2005; Kilcher et al., 2010). The observed NLIWs are associated with SSCs from 5 to 130 mg l^{-1} (Figs. 4 and 6). Thus, the modern Columbia River continues to influence canyon sediment transport by generating NLIWs which travel over the shelf and canyon head. These findings highlight the role of the Columbia River NLIWs as a short-lived yet regularly occurring process that can act to resuspend sediment, transport it shoreward, and inhibit deposit consolidation.

5.2. Canyon heads as down-canyon sediment sources

Despite multiple processes acting to promote deposition within the uppermost head of Astoria Canyon during summer, long-term accumulation is greater in deeper regions of this study site (Fig. 9). Time series observations indicate that fine-grained sediment is deposited and retained in the canyon head over the course of a summer driven by brief upslope currents associated with internal waves and landward flow driven by canyon-confined coastal upwelling. Sediment core data, however, did not constrain the zone of most rapid modern accumulation along the canyon thalweg, as accumulation rates consistently increased to 500 m water depth, the downcanyon limit of sampling. While accumulation rates are consistent with other studies in this area (Nittrouer et al., 1979; Carpenter et al., 1982), the fine spatial resolution presented here highlights this progressive increase between 150 and 500 m water depth. This apparent contradiction between seasonal time series and century-scale accumulation suggests that the canyon head is a zone of temporary deposition and ultimately sediment export, i.e., that winter processes are capable of flushing sediment to depth.

The balance between sediment accumulation and export in many highstand canyons depends primarily on the balance between sediment deposition and wave-associated shear stress. Analysis of regional OOI data indicates that the largest storms of each year can resuspend the full

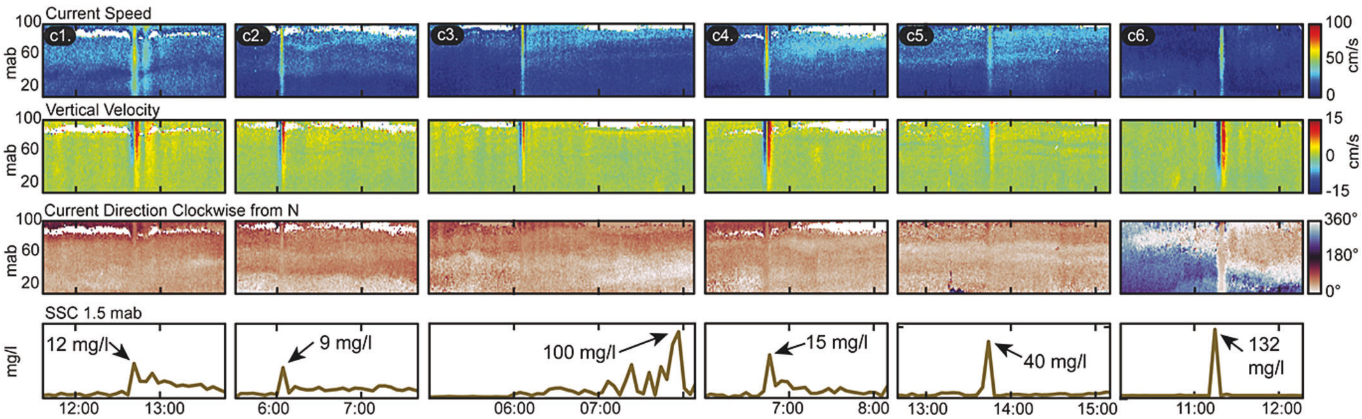

Fig. 4. Expansion of time periods c1 through c6 in Fig. 2 depicting NLIWs. Top row shows elevated current velocities over a period of 5–10 min. Second row depicts vertical velocity. Note that NLIWs begin with negative (downward) vertical velocity. Third row shows that current directions are all moving to the east at a heading of ~90° from N. Bottom row shows increased SSC during and after NLIW passage. Note: concentration scales vary for each event and are defined by the peak annotated.

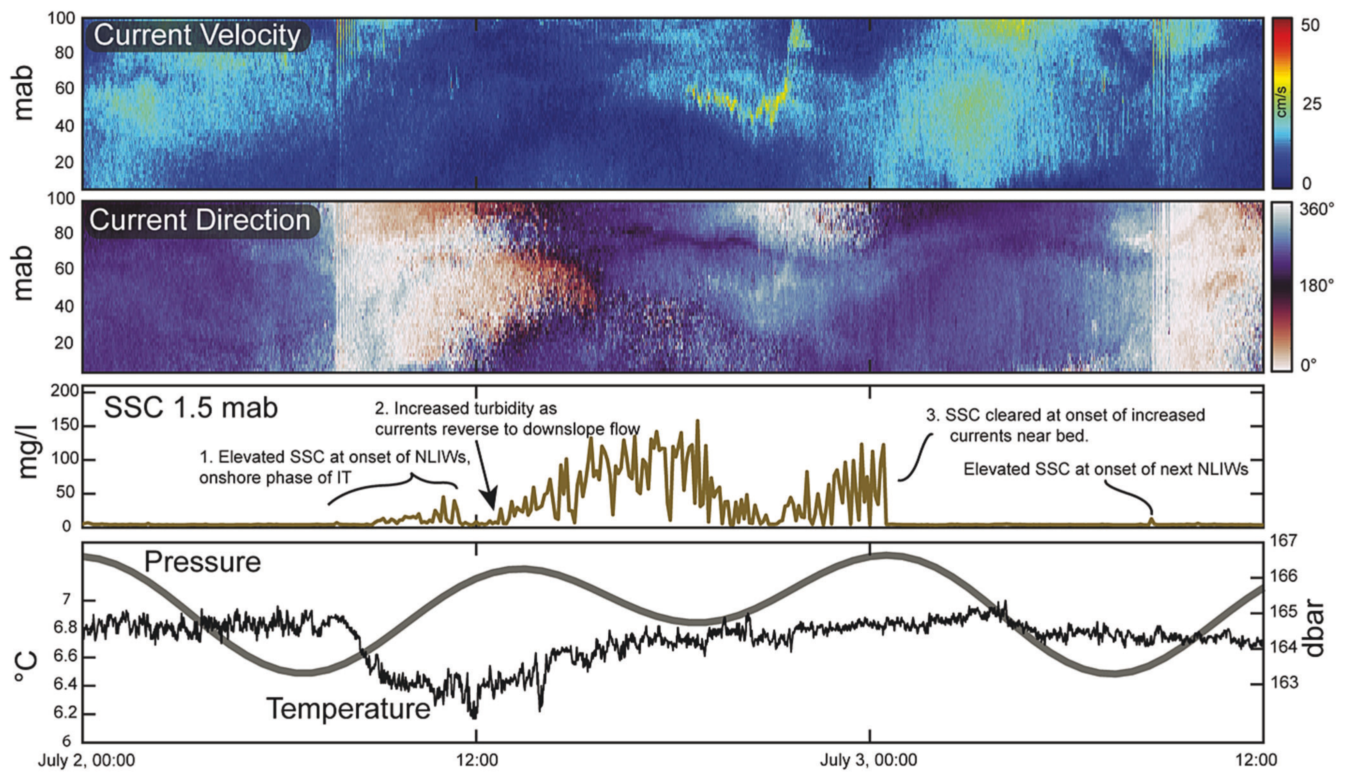

Fig. 5. Expansion of time period d in Fig. 2, showing conditions during elevated suspended sediment concentrations over a 36-h period. The onshore phase of the internal tide arrives at approximately 07:00, accompanied by a sharp drop in temperature and slightly elevated SSC. The large increase in SSC was associated with warming temperatures and downslope flow after the passage of the bore.

grain size distribution of sediment delivered to the canyon head during summer (Fig. 1; Fig. 3). Many of the large storms along the Cascadia Margin are low-pressure systems which drive southerly winds over the shelf and slope, synchronously affecting several canyon systems. In this region, large waves are typically coherent with southerly winds, increasing both bed stress and the capacity for downslope sediment flux. While the precise winter sediment transport mechanisms remain unknown, it is likely that Astoria Canyon exhibits similar attributes of winter-storm-driven transport to the nearby Cascadia Margin canyons: Quinault, Barkley, and Eel Canyons (Baker and Hickey, 1986; Puig et al., 2004; Mullenbach et al., 2004; Lamb et al., 2008; Thomsen et al., 2017).

Among the range of plausible transport mechanisms, prior studies have demonstrated the primary importance of nepheloid layers for delivering sediment into canyon systems during storms (Cutshall et al., 1986; Baker and Hickey, 1986; Snyder and Carson, 1986; Sternberg, 1986), and our findings are consistent with these studies. Storm-driven downcanyon flows are another likely mechanism which transports wave resuspended sediment from the canyon head, as canyon axis currents exhibit a strong response to downwelling-favorable wind stresses (Fig. 2). Last, while less common, the possibility for downslope transport via sediment gravity flows also exists for Astoria Canyon. SSC observations greater than 150 mg l^{-1} were achieved in the summer time series (Fig. 2), and lesser concentrations are known to trigger density-driven flow (Puig et al., 2013; Hage et al., 2019). One sediment gravity flow deposit was collected from the Astoria Canyon rim (140 m water depth, Fig. 9, core 22), and longer cores from upper Astoria Canyon (>500 m,

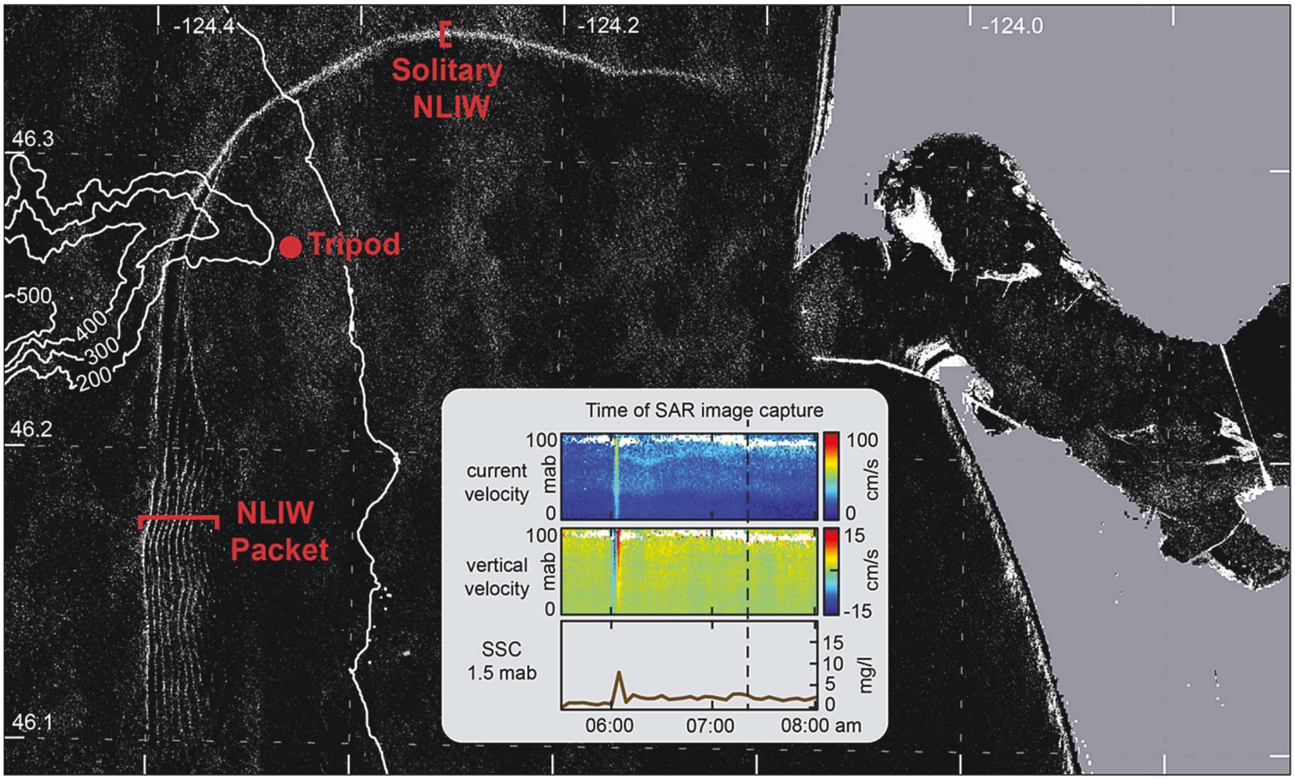

Fig. 6. Synthetic Aperture Radar (SAR) image and contemporaneous time series data showing Columbia River plume-associated NLIWs traveling over Astoria Canyon during the tripod deployment. Main panel: SAR image taken at 07:21 AM on May 31, 2019 with plume-associated NLIWs. Inset: concurrent ADCP and OBS data collected at the tripod site, documenting the passage of a NLIW of depression shortly before the SAR image was taken. Satellite data from Copernicus Sentinel-1 and the European Space Agency.

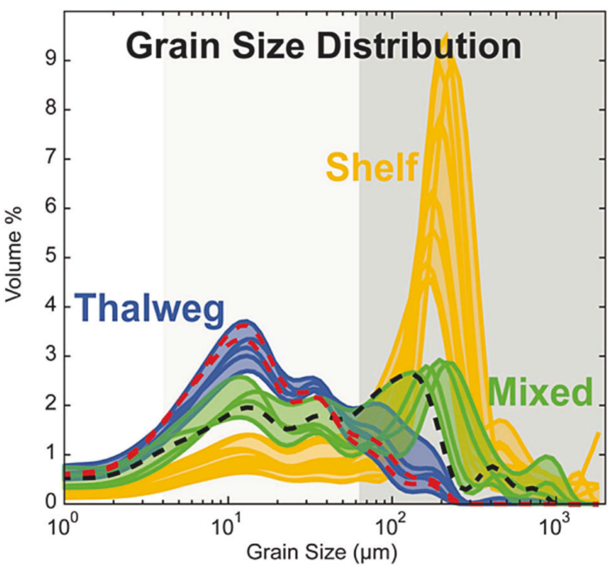

Fig. 7. Grain size distributions of seafloor sediment, colored by environment. Dashed lines are the Astoria Canyon tripod seabed sample (black) and tripod sediment traps (red). Gray shading corresponds to clay, silt, and sand size fraction. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Carlson, 1967) indicate modern sediment gravity flow activity in this region. These events do not represent the dominant mode of sediment accumulation in this canyon, as they are short-lived and relatively unpredictable events. Such flows are challenging to capture with in-situ deployments, and modeling studies may be useful in quantifying the importance of these events in the total canyon sediment budget.

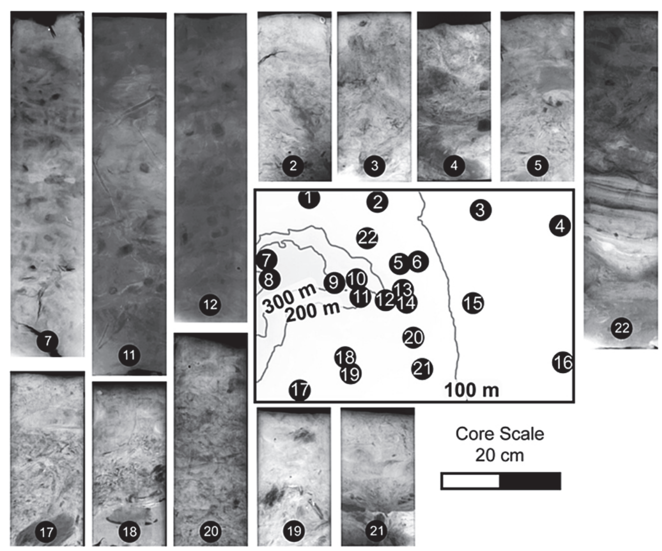

Fig. 8. Selected sediment core x-radiographs depicting internal density structure. Darker shades indicate lower bulk density. Biogenic structures are dominant across the study site, with the exception of region where core 22 was collected on the northern shelf rim.

6. Conclusions

The sedimentological and hydrodynamic observations recorded at the head of Astoria Canyon during summer 2019 combined with

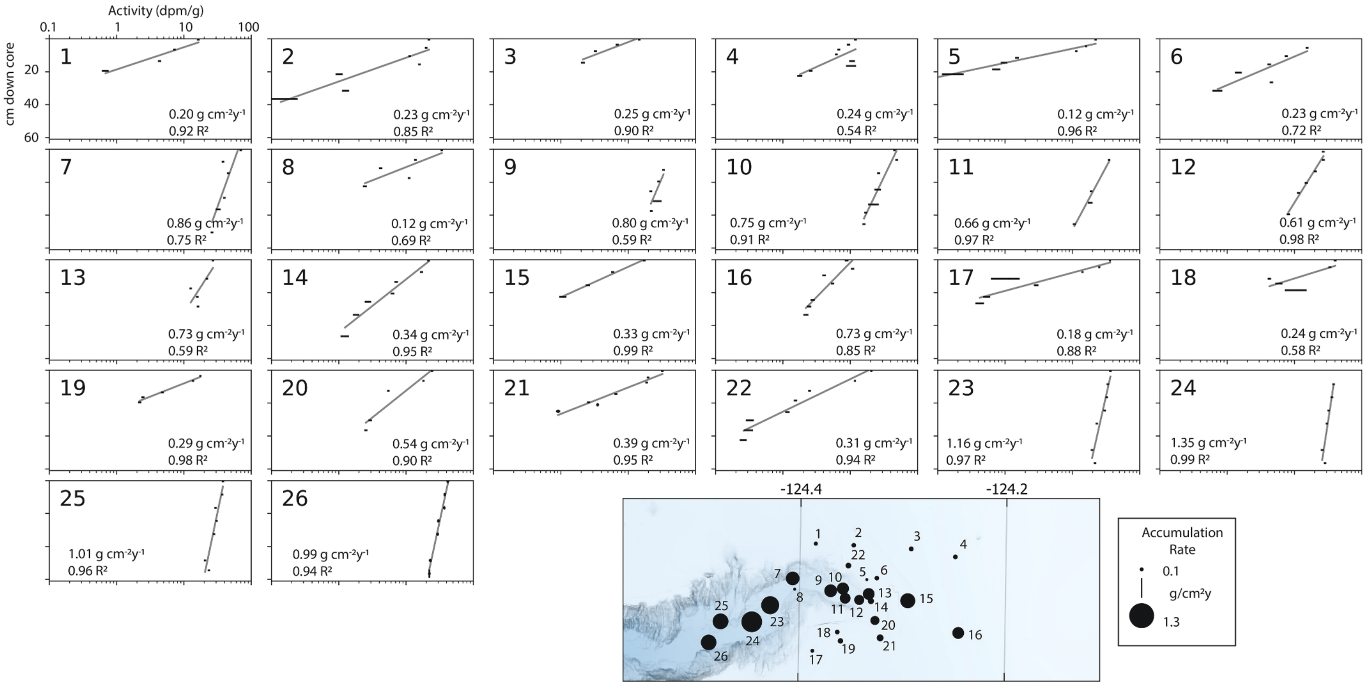

Fig. 9. Excess ^{210}Pb profiles along the head of Astoria Canyon and on the surrounding shelf rim, annotated with their accumulation rate and residual fit. Marker size on the inset map is scaled to accumulation rate. Axis limits in all panels are equal and labeled on the upper left panel.

interannual wind and wave data provide insight into how highstand canyons store and transport sediment under modern conditions. Oceanic processes continue to resuspend and route sediment through the shelf-incising canyon head, supporting the concept that even infilling canyon heads remain energetic sites of sediment transport. Sediment transport during summer is driven by internal tides and river plume NLIWs under upwelling conditions, and these processes provide an environment that seasonally loads canyon heads with unconsolidated fine-grained sediment. During winter, the canyon head is subject to large swell and downwelling wind conditions, driving shear stresses that are sufficient to resuspend all seabed sediment for multiple days per year. These conditions can be found in many relatively quiescent canyons on temperate margins globally. As a result, peak rates of sediment accumulation are focused deeper than the canyon head, indicating canyon head sediment export during sea level highstand.

Submarine canyons throughout the world are often placed on a spectrum between active, energetic sites of sediment flushing and erosion, or quiescent, infilling canyons with minimal transport events. Relative to other canyons globally, Astoria Canyon exhibits traits closer to the quiescent endmember, which is broadly representative of the majority of canyons today. Even so, data from Astoria Canyon suggests that hydrodynamics remain capable of periodic sediment resuspension and transport, and modern stratigraphy from the canyon head suggests sediment gravity flow events do occur, however infrequent. This work demonstrates that multiple processes remain active in canyon systems during sea level highstand and suggests that other quiescent highstand canyons also may be capable of continued sediment export and density-driven flow triggering.

CRediT authorship contribution statement

E.J. Lahr: Writing – review & editing, Methodology, Investigation. **A.S. Ogston:** Writing – review & editing, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **J. C. Hill:** Writing – review & editing. **H.E. Glover:** Writing – review & editing. **K.J. Rosenberger:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be accessed via the Dryad data repository at https://doi.org/10.1016/j.margeo.2025.107516.

Data availability

The data supporting the findings of this study are publicly available via the Dryad repository at DOI https://doi.org/10.5061/dryad.3ffbg79tc\. The dataset includes all relevant time series and sedimentological data and can be accessed and reused in accordance with the repository's guidelines.

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